

Basic Network Discovery

Task: Authorized Local Network Discovery

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Target Environment: Metasploitable 2

Scanning Platform: Kali Linux

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A screenshot of a Kali Linux terminal window. The window title is 'Firefox ESR'. The terminal shows a user prompt '(kali@kali)~' followed by a command 'ping 192.168.31.4'. The output shows a successful ping to 192.168.31.4 with 7 packets transmitted and 7 received, 0% packet loss, and a time of 6049ms. The user then enters a second command 'ping -c 4 192.168.31.4', which also shows a successful ping with 4 packets transmitted and 4 received, 0% packet loss, and a time of 3035ms. The terminal window has a dark background with a Kali Linux logo watermark.

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1). Introduction

This report documents the Day 2 cybersecurity internship task, which focuses on basic network discovery in an authorized local laboratory environment. The lab was created using Kali Linux as the scanning machine and Metasploitable 2 as the intentionally vulnerable target machine.

The purpose of the exercise was to understand the basic process of identifying a target system, verifying network connectivity, scanning for open ports, and identifying the services running on those ports using Nmap.

2). Objective.

The objectives of this exercise were:

- To create an authorized local cybersecurity testing environment.
- To identify the IP address of the target machine.
- To verify connectivity between the Kali Linux and Metasploitable machines.
- To perform a basic Nmap network scan.
- To identify open ports and associated network services.
- To document the findings using screenshots and tables.

3). Lab environment.

Component	Details
Scanner	Kali Linux
Target	Metasploitable 2
Virtualization	Virtualbox
Network type	Host-Only Network
Scanning tool	Nmap
Purpose	Authorized security testing

4). Methodology.

4.1). Network configuration.

Both Kali Linux and Metasploitable 2 were configured within the same isolated Host-Only virtual network. This configuration allowed the machines to communicate with each other without exposing the intentionally vulnerable target directly to the external network.

The environment was used only for authorized cybersecurity testing.

4.2). Ip address identification.

```
4: eth2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:07:43:23 brd ff:ff:ff:ff:ff:ff
    inet 192.168.31.4/24 brd 192.168.31.255 scope global dynamic noprefixroute eth2
        valid_lft 300sec preferred_lft 300sec
    inet6 fe80::a00:27ff:fed0:ff2f/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

```
(kali@kali):[~]
```

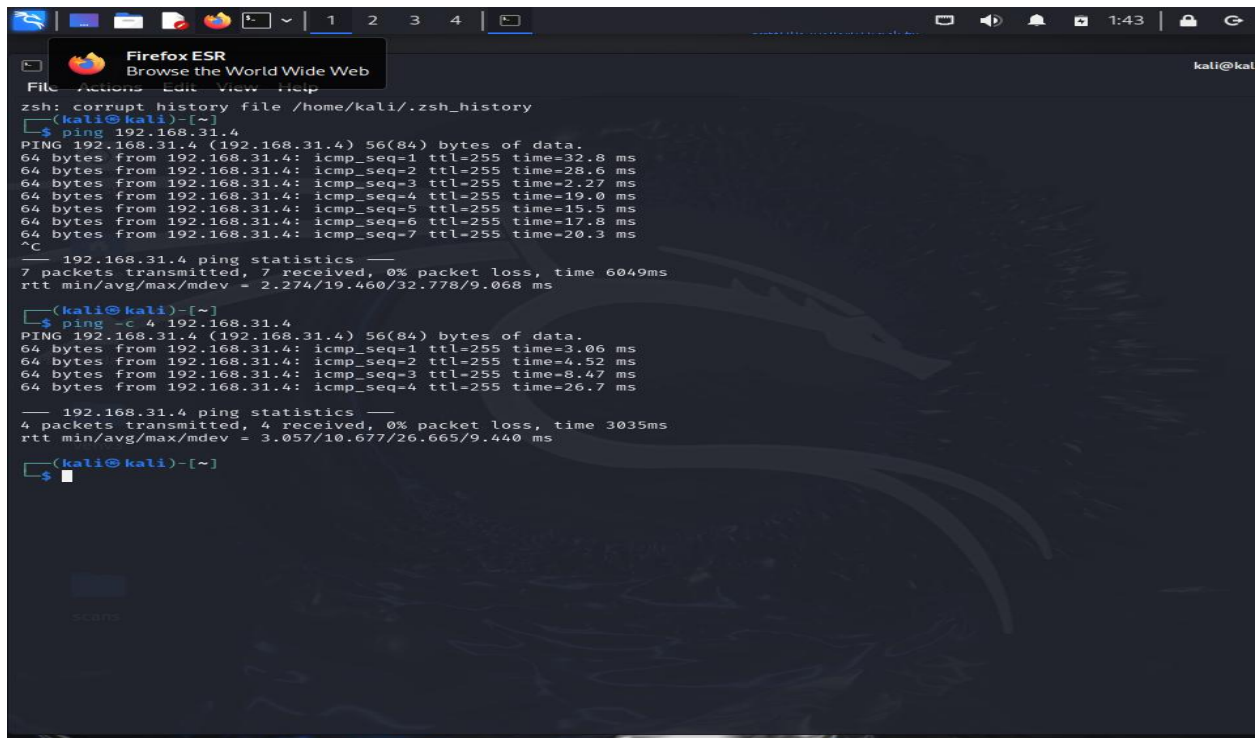
This is the host-only virtual network ip address.

```
link/ether 08:00:27:d0:ff:2f brd ff:ff:ff:ff:ff:ff
inet 192.168.31.4/24 brd 192.168.31.255 scope global eth0
inet6 fe80::a00:27ff:fed0:ff2f/64 scope link
    valid_lft forever preferred_lft forever
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:d0:ff:2f
          inet addr:192.168.31.4  Bcast:192.168.31.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fed0:ff2f/64  Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:11 errors:0 dropped:0 overruns:0 frame:0
          TX packets:59 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:3720 (3.6 KB)  TX bytes:9118 (8.9 KB)
          Base address:0xd020  Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128  Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:142 errors:0 dropped:0 overruns:0 frame:0
          TX packets:142 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:42429 (41.4 KB)  TX bytes:42429 (41.4 KB)
msfadmin@metasploitable:~$ _
```

This is the virtual host-only virtual ip address which is target ip address. In this lab metasploitable vm has been used as the target. Which will be used for connectivity test and Nmap scan.

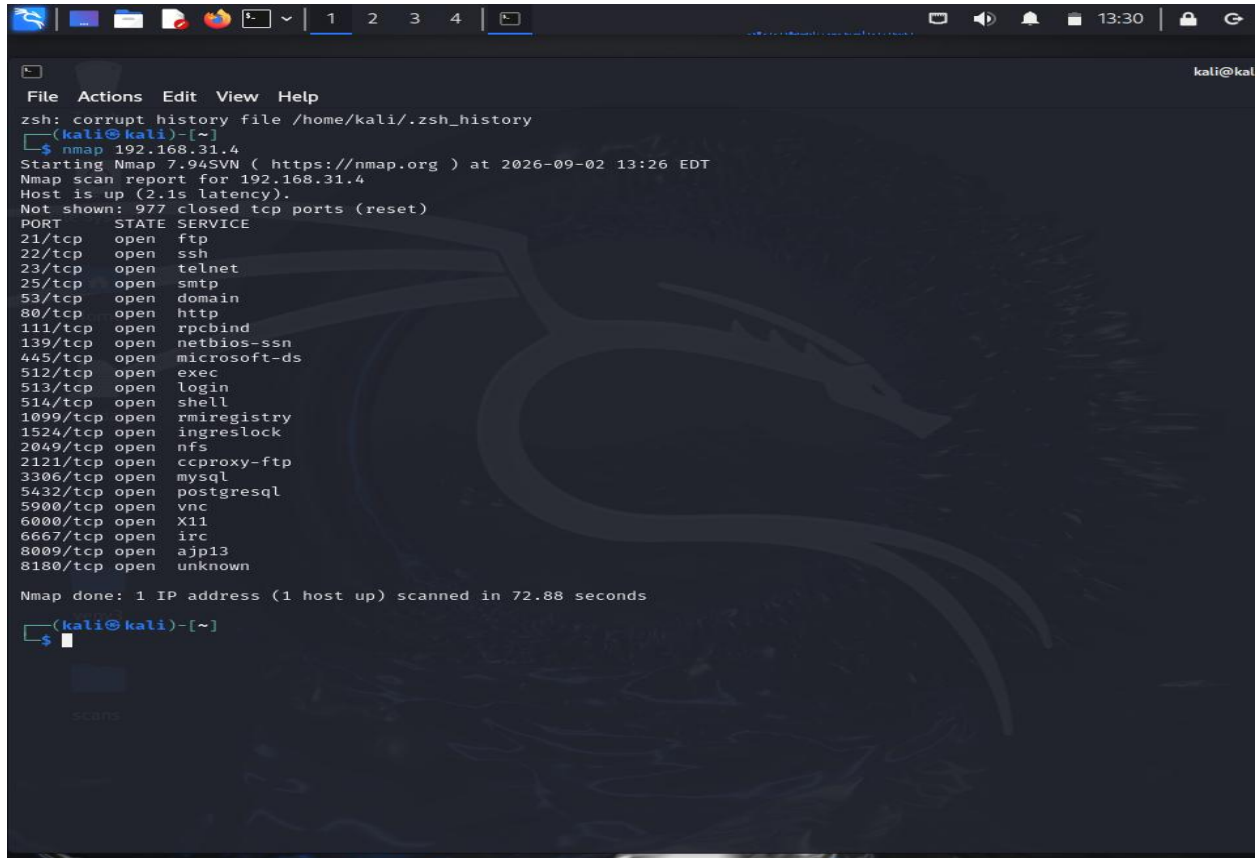
4.3). Connectivity verification.

A screenshot of a Kali Linux terminal window. The terminal shows a series of commands and their outputs. At the top, there's a message about a corrupt history file. The user runs a standard ping command to 192.168.31.4, which shows 7 successful ICMP packets with varying response times. Then, the user runs a continuous ping command with a count of 4, which shows 4 successful ICMP packets. The terminal output is as follows:

```
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
$ ping 192.168.31.4
PING 192.168.31.4 (192.168.31.4) 56(84) bytes of data:
64 bytes from 192.168.31.4: icmp_seq=1 ttl=255 time=32.8 ms
64 bytes from 192.168.31.4: icmp_seq=2 ttl=255 time=28.6 ms
64 bytes from 192.168.31.4: icmp_seq=3 ttl=255 time=2.27 ms
64 bytes from 192.168.31.4: icmp_seq=4 ttl=255 time=19.0 ms
64 bytes from 192.168.31.4: icmp_seq=5 ttl=255 time=15.5 ms
64 bytes from 192.168.31.4: icmp_seq=6 ttl=255 time=17.8 ms
64 bytes from 192.168.31.4: icmp_seq=7 ttl=255 time=20.3 ms
^C
 192.168.31.4 ping statistics:
 7 packets transmitted, 7 received, 0% packet loss, time 6049ms
 rtt min/avg/max/mdev = 2.274/19.460/32.778/9.068 ms
(kali@kali)-[~]
$ ping -c 4 192.168.31.4
PING 192.168.31.4 (192.168.31.4) 56(84) bytes of data:
64 bytes from 192.168.31.4: icmp_seq=1 ttl=255 time=3.06 ms
64 bytes from 192.168.31.4: icmp_seq=2 ttl=255 time=4.52 ms
64 bytes from 192.168.31.4: icmp_seq=3 ttl=255 time=8.47 ms
64 bytes from 192.168.31.4: icmp_seq=4 ttl=255 time=26.7 ms
^C
 192.168.31.4 ping statistics:
 4 packets transmitted, 4 received, 0% packet loss, time 3035ms
 rtt min/avg/max/mdev = 3.057/10.677/26.665/9.440 ms
(kali@kali)-[~]
$
```

Connectivity between the Kali Linux scanning machine and the Metasploitable target was verified using the ping command. Four ICMP packets were sent to the target, and successful responses were received. This confirmed that the two virtual machines could communicate over the configured network.

4.4). Nmap scanning.



```
File Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
$ nmap 192.168.31.4
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-02 13:26 EDT
Nmap scan report for 192.168.31.4
Host is up (2.1s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown

Nmap done: 1 IP address (1 host up) scanned in 72.88 seconds
(kali@kali)-[~]
$
```

Nmap was used to perform a basic TCP port scan against the authorized Metasploitable target. The scan was performed to identify accessible ports and the services associated with those ports.

A service and version detection scan was subsequently performed using the `-sV` option to obtain additional information about the services running on the discovered ports.

5). Results

5.1). Target information.

Information	Result
Target	Metasploitable 2
Target Ip	192.168.31.4
Host status	Up
Scanner	Kali linux.

5.2). Discovered services.

Port	Protocol	State	Service
21	TCP	Open	FTP
22	TCP	Open	SSH
23	TCP	Open	Telnet
25	TCP	Open	SMTP
53	TCP	Open	DNS
80	TCP	Open	HTTP
445	TCP	Open	SMB

3306	TCP	Open	MySQL
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6). Conclusion.

The Day 2 network discovery exercise successfully demonstrated the basic process of performing reconnaissance within an authorized local laboratory environment. Kali Linux was used to identify and communicate with the Metasploitable target, and Nmap was used to discover open ports and network services.

The exercise provided practical experience in host identification, connectivity verification, port scanning, service enumeration, and security documentation. The discovered services can be used as a starting point for further authorized vulnerability assessment and security testing.